

Relevance of the Hydrolysis State of Silyl Acrylate-Based Coatings on Their Antifouling Performance

Louisa Vogler, Regina Kopecz, Onur Özcan, Kelli Hunsucker, Cierra Braga, Tim Patschorke, Dylan Thomas Eggers, Geoffrey Swain, and Axel Rosenhahn*



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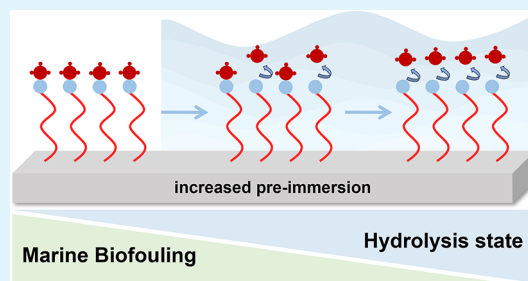
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ABSTRACT: Silyl (meth)acrylates represent indispensable building blocks for the synthesis of self-polishing copolymers (SPCs), with triisopropylsilyl acrylate (TiPSA) being one of the most important monomers for the fabrication of such coatings. Polymer coatings were synthesized that are based solely on TiPSA via a grafting-through polymerization, and their in situ hydrolysis behavior has been analyzed. For the in situ characterization, captive bubble contact angle (CBCA) measurements were conducted, and the mechanical properties of the different hydrolysis states were analyzed using pendulum hardness measurements. Moreover, the capability of the coatings to absorb water and their stability were gravimetrically investigated using immersion experiments in aqueous media. Based on the results, selected hydrolysis states were challenged against marine organisms using both laboratory and short-term field exposure in the ocean, where a connection between the hydrolysis state and the respective antifouling performance of the coatings could be established.

KEYWORDS: TiPSA, self-polishing copolymer, methacrylate, contact angle, marine biofouling, fouling release, *Amphora*, field test



INTRODUCTION

The accumulation of biological matter on submerged, artificial surfaces is a massive problem within the shipping industry that causes detrimental ecological and economic impacts.¹ Such undesirable biofouling results in an increased drag directly leading to higher fuel consumption and an increased release of polluting substances (e.g., CO₂ or NO_x).² Apart from that, the invasion of neobiota through cross-contamination displays a menacing problem for indigenous species.³ Strategies to combat the undesired biomass deposition frequently rely on the use of biocidal antifouling (AF) coatings.^{4–6} As particularly promising appears the combination with ultralow fouling chemistries such as silicones,^{7–10} amphiphilic copolymers^{11,12} or zwitterion containing polymers.^{13,14} Also, superhydrophobic composite coatings,^{15,16} which reduce fouling via a physical protection layer (plastron), are promising candidates for the protection of underwater structures. By combining different materials and adapting various designs given by nature, a broad collection of novel, bioinspired coatings can be fabricated.¹⁷ Besides the self-cleaning lotus leaf,¹⁸ or the slippery *Nepenthes* pitcher plant,¹⁹ fish and amphibians also utilize a specific mechanism to adapt to environmental influences.²⁰ Mucus production by fish results in a protective surface layer, composed of hydrophilic biopolymers such as glycoproteins.^{20,21} Thus, the production of mucus equips these aquatic organisms with a self-renewing surface that minimizes the interaction between the organism and its surrounding

environment.^{20,22} In contrast, the pilot whale *Globicephala melas* uses a combination of chemical and physical properties to prevent fouling. The skin of *G. melas* is characterized by nanometer-sized pores with a diameter of approximately 0.1–1.2 μm²³ that are enclosed by nanoridges and allow the secretion of a sacrificial zymogel. Through the jumping motion of the pilot whale, the detachment of fouling organisms from these smooth surfaces is favored.^{24,25} In addition to this sacrificial effect, the zymogel contains enzymes that enable the degradation of marine organisms' secreted adhesives by hydrolysis.²⁶ For crustose coralline algae (CCA), sloughing of epithelial cells plays a crucial role in fouling prevention.²⁷ In a previous study, species-specific mechanisms for the prevention of fouling were described in three various CCAs. Investigations with scanning electron microscopy (SEM) revealed the shedding of an entire epithelial cell layer for *Sporolithon ptychoides* and *Neogoniolithon fosliei*, while *Hydrolythion onkodes* sheds only single epithelial cells. For all three species, the respective dead organisms were completely overgrown after 2 weeks compared to the intact organisms,

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highlighting the importance of epithelial self-renewal for antifouling management via continuous self-shedding of attached fouling organisms of such organisms.²⁸

Compared to these self-renewing and self-shedding systems found in nature, modern self-polishing copolymers (SPCs) can be considered as a bioinspired synthetic analogue. Since the ban on membrane-active tributyltin (TBT) in SPCs, trialkylsilyl (meth)acrylate components have attracted increasing attention due to their good compatibility with several alternative biocides.²⁹ Nowadays, triisopropylsilyl acrylate (TiPSA) and triisopropylsilyl methacrylate (TiPSMA) are some of the most important hydrolyzable monomers found in commercially available silyl ester copolymers, with contents accounting for more than 50%.^{30–33} Both components carry a hydrolyzable triisopropylsilyl pendant group, allowing a controlled surface erosion over time in seawater due to the controlled surface transformation from hydrophobic to more hydrophilic by ester hydrolysis.³⁴ Such wettability changes were formerly assessed by contact angle goniometry, and for silyl ester copolymers, a rapid decrease was reported within the first days of immersion. Interestingly, it was observed that the contact angles equilibrated after 7–30 days of immersion, depending on the copolymer composition.^{34–37} Previous studies have also shown that prehydrolyzed SPCs show improved antifouling performance when compared to the pristine coatings, although the magnitude of this effect differs depending on the tested organism.^{34,38} Dai et al. copolymerized methyl methacrylate (MMA) and TiPSA (85/15) via radical polymerization and showed that the slow hydrolysis kinetics lead to a slight decrease of the contact angle, and several days of immersion are required until the contact angles equilibrate. After 3 days of preimmersion, they were able to observe a 39% decrease in the *Navicula incerta* density compared to the respective nonleached coatings, while the density of the bacterium *Pseudomonas* sp. remained nearly unaffected.³⁴

Literature provided first indications that the hydrolysis state of the silyl (meth)acrylate compounds in the copolymer coatings may affect the antifouling performance of the coatings. Until now, an in-depth understanding of coatings consisting solely of silyl acrylate building blocks, their in situ hydrolysis behavior at the coating-water interface, and the effect on the colonization by different fouling organisms under laboratory and field conditions is still lacking. Such a study includes the crucial question of whether the antifouling performance can be significantly improved by a control of the hydrolysis state of the surface when in contact with an aqueous medium. We performed a fundamental study on the time-dependent hydrolysis states of TiPSA coatings and tested their influence on fouling retention under laboratory and field conditions to cover a variety of marine organisms. For this purpose, we fabricated a nonablative TiPSA-based polymer, allowing us to investigate the interplay between water and the external surface layer. Coatings were prepared via a photoinduced, grafting-through free-radical polymerization according to previously published protocols.³⁹ A cross-linker was used to inhibit any ablation of polymer chains of the coating and restrict the study to the hydrolysis state of the TiPSA compound. After curing, polymerization success and characterization of the hydrolytic behavior of the coatings were assessed by ATR-FTIR and static water, as well as captive bubble contact angle goniometry, respectively. Finally, the different preimmersed polymers were challenged against marine fouling in both laboratory (*Amphora*

sp. and *Cobetia marina*) and compared against short-term field immersion experiments.

EXPERIMENTAL SECTION

Chemicals

3-(trimethoxysilyl)propyl methacrylate (MPTMS), acetonitrile, ethylene glycol diacrylate (EGDA), 2,2-dimethoxy-2-phenylacetophenone (DMPAP), and poly(butyl methacrylate) (pBMA) were purchased from Sigma-Aldrich. Acetone, ethanol p.a., and 2,2,2-trifluoroethanol (TFE) were received from vwr. Triisopropylsilyl acrylate (TiPSA) was purchased from abcr. Glutaraldehyde was received from Alfa Aesar. pBMA containing 1 mol % of cross-linker 2-(4'-benzoylphenoxy)-ethyl methacrylate (BPEMA) was kindly provided by the group of Prof. Laschewsky, University of Potsdam. All chemicals were used as received without further purification.

Preparation of Poly(TiPSA)-Based Coatings

Synthesis of the polymer coatings was accomplished by a photoinduced grafting-through polymerization approach.³⁹ First, the glass substrate (Marienfeld, Lauda-Königshofen, Germany) was functionalized with MPTMS according to a previously published protocol.⁴⁰ In short, the substrate was activated in an oxygen plasma (0.4 mbar, 80 W, 22 kHz, miniFlecto-PC-MFC, Gala Instruments, Germany) for 20 min, which resulted in the formation of free hydroxyl groups on the surface, followed by subsequent immersion in a solution of 1% (v/v) MPTMS in acetone for 90 min. Afterward, the substrate was sonicated in acetone and EtOH for 5 min each before drying in a stream of nitrogen. For the coatings, TiPSA was used as hydrolyzable monomer unit, and 0.32 wt % (0.46 mol %) EGDA (EGDA/monomer) were added as cross-linker along with 1 wt % (0.9 mol %) DMPAP (DMPAP/monomer) as photoinitiator. The transparent mixture was vortexed and degassed for 30 min under the exclusion of light. Afterward, the HybriWell reaction chamber (SecureSeal adhesive chamber, chamber volume: 275–320 μ L, size $L \times W \times$ thickness: 60 mm $\times$ 21 mm $\times$ 0.15 mm, ports diam.: 3.2 mm; Grace Bio-Laboratories) was filled with the monomer solution before being irradiated for 10 min by an UV source (UVAcube 100, Dr. Hönle AG, 100 W iron-doped mercury vapor lamp, cutoff filter >310 nm, source-sample distance: 18 cm, 15 mW/cm²). After 16 h of airtight curing, the reaction chamber was detached from the glass substrate, and the silyl acrylate coatings were stored under argon until further use.

Preparation of Hydrophobic pBMA

Fabrication of the negative control for the laboratory organism experiments was performed by using hydrophobic pBMA containing 1 mol % of cross-linker BPEMA. After dissolving 1 wt % (w/w) of the pBMA in TFE, the solution was spin-coated (WS-650MZ-23NPP/Lite, Laurell Technologies Corporation, 15 s at 100 rpm, followed by 30 s at 3000 rpm) on an MPTMS-functionalized glass substrate. The polymers were cured by UV irradiation for 30 min.⁴¹ Negative control for the short-term field experiments was prepared by spin-coating a solution of 1.4 wt % (w/v) pBMA in ethyl acetate on an MPTMS-functionalized glass substrate.

Water Contact Angle Goniometry

Wettability of the coatings was assessed by two different methods, sessile water contact angle (WCA) and captive bubble contact angle goniometry (CBCA), respectively. For the WCA, a water droplet (20 μ L) was placed on the polymer coating using a custom-built goniometer. The droplet was recorded with a charge-coupled device (CCD) camera, and the contact angle was fitted according to Young's equation. For the CBCA, the coating was submerged upside down in a salt water-filled (SW, seven most abundant salts from seawater⁴²), transparent container before an air bubble was fixed on the coating from underneath with a syringe. The bubble was recorded with a CCD camera, and evaluation was performed by using the angle tool of ImageJ. The coatings were investigated over a period of 7–14 days. For all samples, a total of three replicates were analyzed.

Preimmersion

To investigate different hydrolysis states, the silyl acrylic coatings were preleached for the desired durations ($t = 0$ h, 3 h, 1 d, 7 d, and 14 d) in SW on an orbital shaker (Neolab, DOS-10L) at 65 rpm. Afterward, the coatings were rinsed with Milli-Q (MQ) water to prevent salt precipitation. The coatings were directly transferred to the required follow-up experiments.

ATR-FTIR

ATR-FITR spectra were recorded by using a Tensor 27 from Bruker (200 scans, 4 cm^{-1} resolution, wavenumber range $4000\text{--}600\text{ cm}^{-1}$) with a VariGATR Ge-ATR crystal (Harrick Scientific Products, USA), with a detection of the FTIR signals by a mercury cadmium telluride (MCT) detector.

Pendulum Hardness

Mechanical properties of the dry, preleached coatings were characterized by measuring the pendulum hardness using a König pendulum hardness tester (BYK, byko-swing König/Persoz). After a deflection of 6° , the time required for the pendulum to reach a swinging amplitude of 3° was determined. The oscillation time on a glass slide was 250 ± 10 s. A total of three replicates were measured for each preleaching time.

Coating Stability Assessment

Stability of the silyl acrylate-based polymers was investigated gravimetrically. The pristine coatings were weighed and immersed for the desired durations in SW under dynamic conditions (65 rpm, orbital shaker). After rinsing with MQ and drying, the coating weight was determined again ($m_{t,\text{dry}}$), and the remaining mass fraction (RMF) was calculated according to eq 1

$$\text{RMF} = \frac{m_{t,\text{dry}}}{m_0} \quad (1)$$

Water Uptake

Water uptake capability of the coatings was gravimetrically determined. After immersion in SW for different durations under dynamic conditions at 65 rpm, the coatings were rinsed with MQ, and excess water droplets were gently removed before the wet coating was weighed ($m_{t,\text{wet}}$). The relative water uptake (RWU) was calculated in relation to the dry weight of the pristine coatings (eq 2)

$$\text{RWU} = \frac{m_{t,\text{wet}}}{m_0} \quad (2)$$

Dynamic Attachment Assay against *Amphora* sp.

Determination of the AF properties was performed by challenging the preimmersed coatings against the marine diatom *Amphora* sp. in a dynamic attachment assay. The samples were directly transferred to the assay after preimmersion in SW to ensure a continuous wetting of the surfaces. For the diatom assay, algae from a continuous culture were used. After resuspension of the diatoms in artificial seawater (ASW, pH 8.0, Instant Ocean) and filtration, the optical density at 444 nm (LAMBDA Bio + Spectrophotometer, PerkinElmer) was measured to adjust the concentration of the diatom solution to 4 Mio cells mL^{-1} . The samples were preincubated in 4 mL ASW for 30 min. Afterward, 4 mL of the diatom solution was added to reach a final concentration of 2 Mio diatoms mL^{-1} , and the coatings were incubated for 60 min on an orbital shaker at 65 rpm. Fixation of the organisms was done with glutaraldehyde at a concentration of 2.5 vol % for 10 min under dynamic conditions. After washing twice with MQ for 10 min each, the samples were dried under the exclusion of light and stored until evaluation.

Dynamic Attachment Assay against *C. marina*

For the dynamic bacterial assay, the original strain from the marine organism *C. marina* was received from the German Collection of Microorganisms and Cell Cultures GmbH (DSMZ, Braunschweig, Germany), and the cultivation of the marine bacterium was performed according to a previously described protocol.⁴³ On sterilized marine

agar plates, cells from frozen aliquots were plated and incubated for 24 h at 18°C . After picking a single colony, 7.5 mL of sterile filtered MB medium was inoculated. Growth of the bacteria was performed overnight at 18°C under dynamic conditions (Neolab, DOS-10L, 70 rpm) until an optical density of $\text{OD}_{600} \geq 1$ was achieved. By filtration and dilution (1:100) of the bacterial suspension, the formation of agglomerates could be prevented. The bacterial solution was further incubated under dynamic conditions until an $\text{OD} = 1$ was reached, as the bacteria are then in the log phase, providing the most reliable results. After centrifugation (Hettich Zentrifugen, Lauenau, Germany) of the culture at 3000 rpm for 5 min, the pellet was resuspended in ASW, and the suspension was adjusted to a concentration of 15 Mio cells mL^{-1} . The bacterial attachment assay was performed according to a previously published protocol.⁴⁴ After preleaching and equilibration for 30 min in 4 mL ASW, the samples were covered with 4 mL of the bacterial solution (15×10^6 cells mL^{-1}) to reach a final concentration of 7.5×10^6 cells mL^{-1} . The samples were incubated with the bacterial solution for 45 min on an orbital shaker at 60 rpm. Afterward, the organisms were fixed using glutaraldehyde (2.5 vol %) for 10 min under dynamic conditions. Before staining the adhered bacteria, the samples were washed twice with MQ for 10 min each at 60 rpm. For staining, the samples were incubated in a solution of propidium iodide (1 mg/mL). Lastly, the coatings were washed with MQ under static conditions and dried under the exclusion of light until further analysis.

Quantification of the Density of Attached Microorganisms

The dynamic attachment assays were evaluated by counting the adhered organisms with a fluorescence microscope (10 $\times$ and 20 $\times$ objective, Texas Red filter set, Nikon Ti-E, Nikon Instruments, Japan). Each sample was automatically scanned using a customized grid with at least 60 fields of view. Counting of the adhered organisms and evaluation was conducted using NIS Elements AR 4.30.02 (Nikon, Germany). The statistical significance of observed differences was analyzed by conducting a one-way ANOVA with a post hoc Tukey analysis (significance level $\alpha = 0.05$).

Sediment Accumulation Assay

The affinity of the preimmersed samples to accumulate silt was investigated spectroscopically. The silt was autoclaved and diluted in seawater until a concentration of 37.5 g/L was reached. The UV–Vis spectra (Agilent Cary 60, Santa Clara, CA) of the pristine coatings were recorded at 440 nm before the samples were immersed in the silt mixture for 60 min on an orbital shaker at 60 rpm. Subsequently, the coatings were washed with SW and MQ for 2 min each to remove nonadhered silt and prevent salt precipitation. After incubation, the UV–Vis spectra were recorded again at 440 nm, and the relative absorption was determined according to eq 3

$$\text{relative absorbance} = \frac{A_{440}}{A_{440,\text{pristine}}} \quad (3)$$

Field Experiments

Besides the laboratory attachment assays, short-term field tests of the preimmersed silyl acrylate-based coatings were carried out at Port Canaveral, Florida ($28^\circ 24' 30.6''\text{N}$ $80^\circ 37' 38.7''\text{W}$) under static and dynamic conditions. For the dynamic tests, the coatings were immersed using a custom-built rotating disk setup at an immersion depth of 0.5 m and a rotation speed of 13 rpm.⁴⁵ In total, the coatings were immersed for 3–4 days, and at least 6 replicates were tested for each coating. Lastly, the samples were fixed with 2.5 vol % glutaraldehyde for 10 min before being washed twice with MQ under static conditions. The dry samples were analyzed using fluorescence microscopy and the fluorescence coverage was determined in least 60 fields of view.

RESULTS

To study the antifouling properties of triisopropylsilyl acrylate (TiPSA) in different hydrolysis states, a cross-linked coating containing solely TiPSA was prepared via free-radical grafting-through polymerization. This approach combines the for-

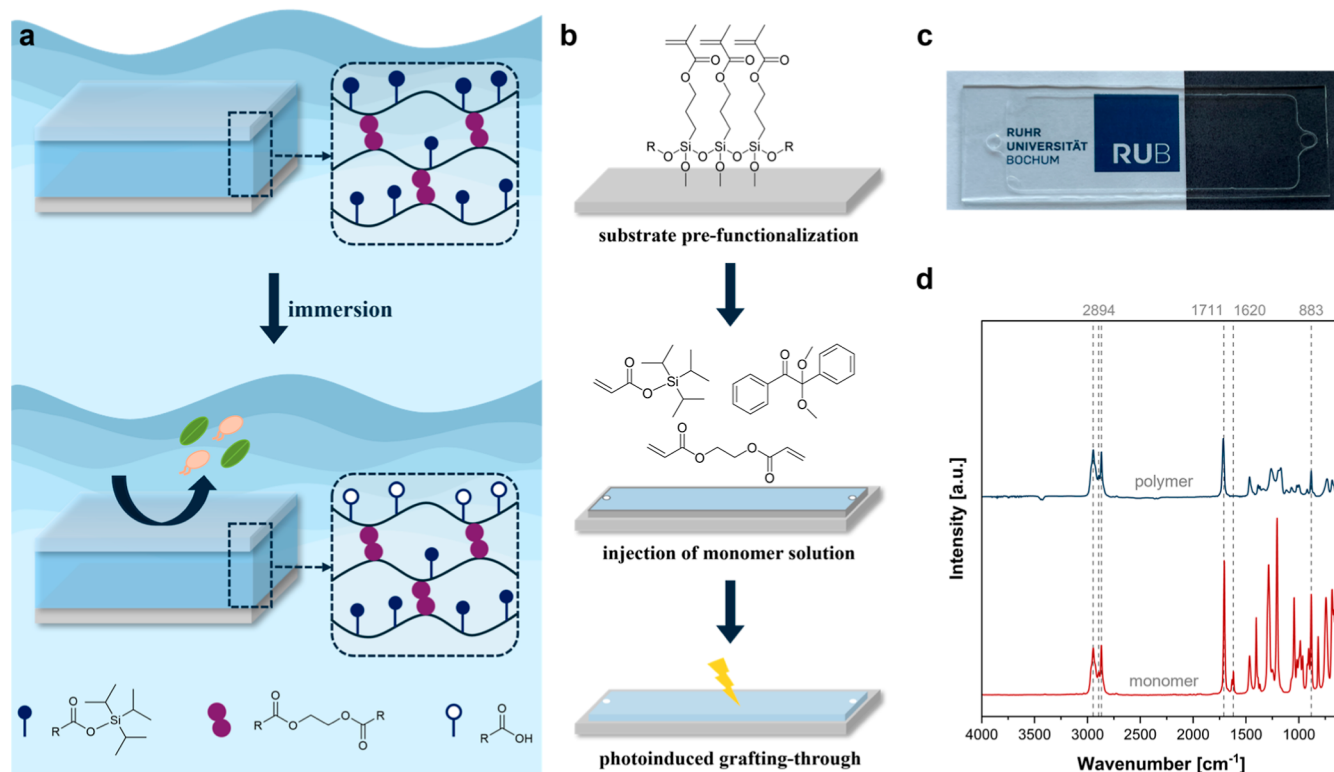


Figure 1. (a) Schematic representation of the coating response to immersion in seawater: after immersion of the silyl acrylic coating, the upper surface layer undergoes hydrolysis, leading to a change in wettability, which ultimately prevents the undesired adhesion of marine organisms. (b) Schematic fabrication of cross-linked p(TiPSA) coatings via grafting-through polymerization. (c) Image of the ultrasmooth, pristine silyl acrylic coating on the glass substrate. (d) ATR-FTIR spectra of the TiPSA monomer (red) and the respective polymer (blue).

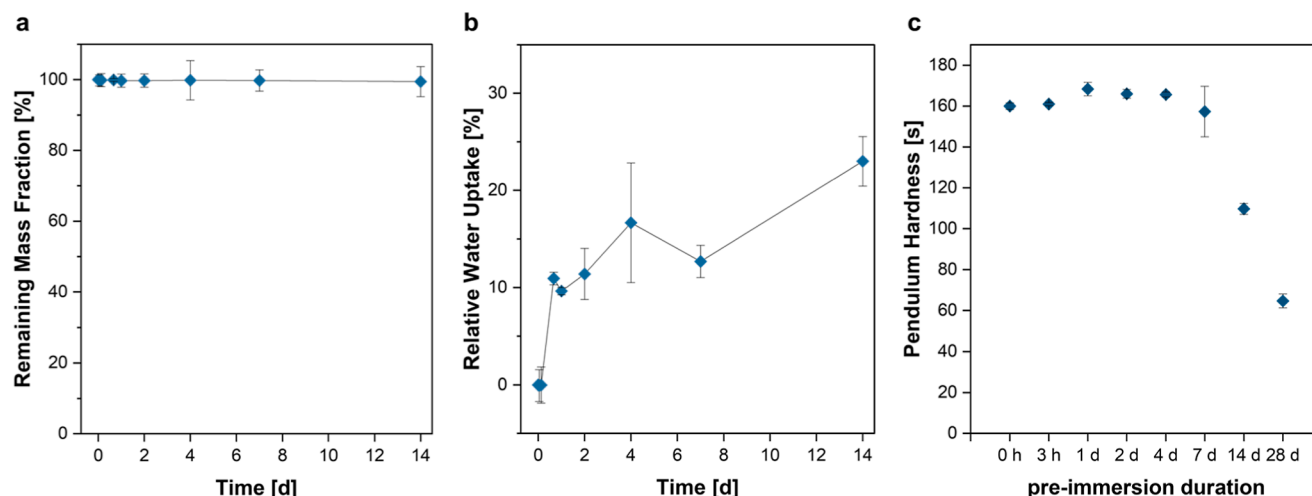


Figure 2. Stability and property changes upon immersion of the coatings in salt water (a) coating stability in SW under dynamic immersion conditions. The coatings were immersed over a period of 14 days at room temperature, and the weight loss was determined after different immersion durations. (b) Water uptake in SW under dynamic conditions over a period of 14 d at room temperature. The relative water uptake was calculated with regard to the pristine coating weight. (c) Pendulum hardness of different hydrolysis states of the dry p(TiPSA) coatings. In all three diagrams, reported values represent the average of three independent replicates. Error bars represent the standard deviation.

mation of macromolecules while simultaneously covalently anchoring the growing polymer chains to the substrate (Figure 1a and b).⁴⁶ The ethylene glycol diacrylate (EGDA) was added as a cross-linker to restrict the study to the hydrolysis state of the silyl acrylate side groups and to avoid any convection of the results with polishing processes, i.e., the shedding of polymer. After UV-initiated polymerization, transparent coatings were obtained (Figure 1c) that were rather hydrophobic

with sessile water contact angles (WCAs) in air of $97 \pm 1^\circ$ and a coating weight and thickness of approximately 130 mg and 150 μm , respectively. The chemical transformations occurring during the UV-induced polymerization process were verified by ATR-FTIR spectroscopy. Spectra of the monomer and the respective pristine silyl acrylate-based coating (Figure 1d) showed peaks at 2946–2868 cm^{-1} and 1711 cm^{-1} that can be ascribed to the C–H stretching vibration and to the carbonyl

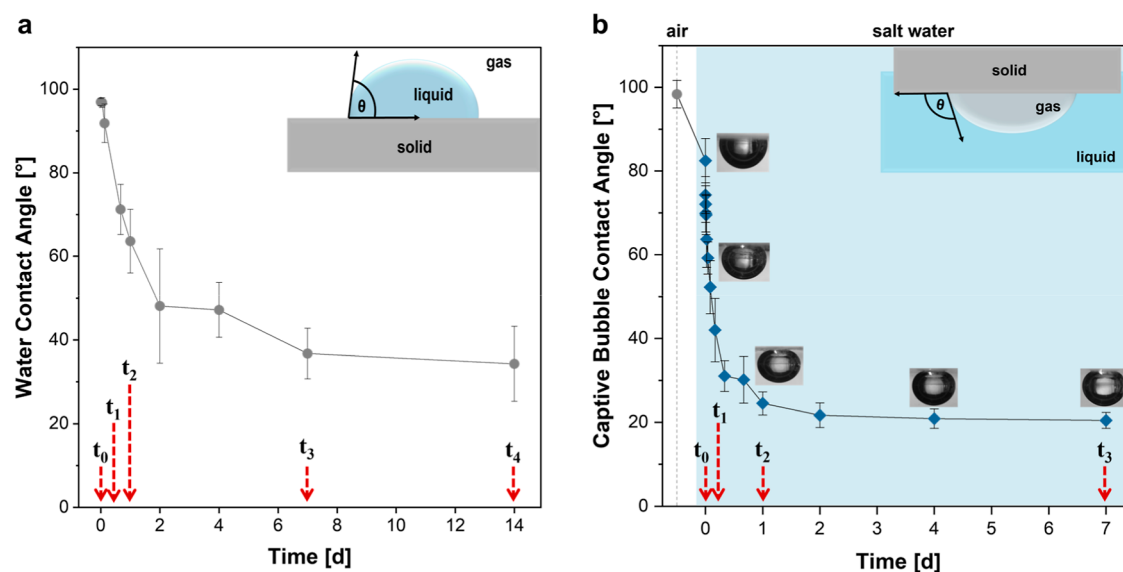


Figure 3. Wettability of the TiPSA coatings (a) static water contact angle of the coatings after incubation in SW at 65 rpm on an orbital shaker before being rinsed with MQ and dried for 24 h at room temperature before measurement. (b) Captive bubble contact angle of the TiPSA-based coating measured under water. For the CBCA, the coatings were incubated in an SW-filled container under static conditions, and the spreading of air bubbles was determined. The abbreviations t_0 – t_4 indicate immersion durations in SW identified as interesting for the biological experiments. All data points are triplicate measurements, and the error bars represent the standard deviation.

group of the ester bond, respectively.⁴⁷ The stretching vibration of the C=C bond at 1620 cm^{-1} for the monomer disappeared nearly completely in the p(TiPSA)-based polymer. The peak at 883 cm^{-1} can be attributed to the stretching vibration of the Si–C bond, which decreased in intensity after polymerization.⁴⁷ The absence of the C=C vibration at 1620 cm^{-1} after polymerization suggests that the vast majority of monomers were successfully converted into the polymer.

Due to the addition of EGDA as cross-linker, the coatings were designed to be stable in seawater to restrict the analysis in this work to the hydrolysis state of the silyl acrylate group and to avoid any convolution with ablative processes. To verify the stability of the coatings under water, the degradation of the coating was investigated over a period of 14 days in SW under dynamic conditions (Figure 2a), and the degradation of the coatings was determined as relative decrease in mass with respect to the pristine polymer film. Over the immersion duration of in total 14 days, the silyl acrylate coating remained stable with >98% with respect to its initial mass. Thus, the cross-linking stabilized the films sufficiently over immersion periods of 2 weeks.

After different immersion durations, the water uptake was determined by the gravimetric analysis of the samples immediately after retrieval (Figure 2b). Within the first 3 h of immersion (time points very close to 0 d), no water uptake was observed for the TiPSA-based coatings. After further immersion, a slight water uptake of 11% and 10% with respect to the initial mass of the coatings was observed for 16 h and 1 d of immersion. From 4 days of immersion on, the increase in water uptake slowly started to converge, and a maximum water uptake of 23% was found after 14 days. Characterization of the mechanical properties of the differently hydrolyzed coatings was done with a pendulum hardness tester (Figure 2c). For the nonleached coatings, a pendulum hardness of 160 ± 2 s was obtained. After an immersion for 3 h, 1 d, and 7 d, nearly no differences in the pendulum hardness were observed for the differently hydrolyzed silyl acrylic coatings. Exceeding

immersion durations of 1 week, the pendulum hardness started to change, and after 14 days of immersion in SW, the pendulum hardness decreased to 110 ± 3 s. The lowest pendulum hardness was observed for the coatings after 28 d of immersion with 65 ± 3 s. For comparison, relatively hard polymers like PEEK have pendulum hardnesses of ~ 220 s, while softer polyurethane elastomers with a modulus of 4.5 MPa have pendulum hardnesses of 50 s.⁴⁸

The time-dependent water uptake and the changes in pendulum hardness during immersion already indicated that changes at the interface occur when the coatings were exposed to seawater. Static water contact angle goniometry was used to assess the wettability in air (Figure 3a), and captive bubble contact angle goniometry was used to monitor the underwater wettability changes in the immersed state (Figure 3b). For the pristine, dry silyl acrylic coating, a WCA of $97 \pm 1^\circ$ was measured. After immersion in SW for 1 h, the WCA remained stable at approximately $97 \pm 1^\circ$, while it decreased to $71 \pm 6^\circ$ and $64 \pm 8^\circ$ after 16 h and 1 d of incubation, respectively (Figure 3a). After 7 days, the WCA was starting to converge, and a change to $37 \pm 6^\circ$ by approximately 60° compared to the dry, pristine coating was observed, which remained nearly constant up to a 14 day immersion with a WCA of $34 \pm 9^\circ$. While the WCA provides information about the wettability of the surface in air, the CBCA is very sensitive to rapid molecular reorientations once the surfaces are immersed in water but also reflects long-term chemical changes at the interface with progressing hydrolysis of the silyl acrylate groups. As shown in Figure 3b, the CBCA decrease occurred much more rapidly than observed for the WCA, and after seconds of immersion, it dropped to $82 \pm 5^\circ$ (up to 15° lower compared to the dry surface, Figure 3a and b). The CBCAs continued to decrease within the first 16 h of immersion to $30 \pm 6^\circ$. After 1 day of incubation in salt water (SW), the CBCA equilibrated at approximately $24 \pm 3^\circ$. Based on the observed changes in wettability (Figure 3), in water uptake (Figure 2b), and the change in pendulum hardness (Figure 2c), interesting

immersion times for the biological investigations of the different hydrolysis states of the p(TiPSA) coatings appeared to be t_0 as the pristine, nonimmersed sample (0 h), t_1 (3 h in SW), t_2 (1 d in SW), t_3 (7 d in SW), and t_4 (14 d in SW) (time points are labeled by the arrows in Figure 3).

In Figure 4, the chemical transformations occurring during immersion and increasing hydrophilization of the silyl acrylic

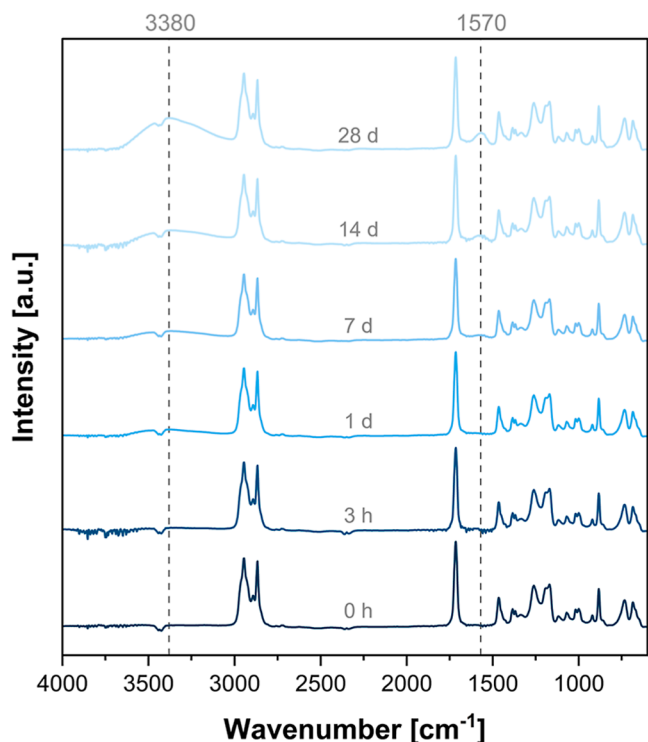


Figure 4. ATR-FTIR spectrum of the p(TiPSA) coatings after different immersion durations in SW.

coating after different immersion durations in SW were studied by ATR-FTIR spectroscopy. The broad peak emerging at 3680–3110 cm^{-1} can be attributed to the OH stretching vibration of the free carboxylic acid groups resulting from the hydrolysis of the triisopropylsilyl groups, and its intensity increased with increasing incubation time in SW.³⁴ In addition, the peak at 1570 cm^{-1} is formed after 14 days of immersion in SW, which can be ascribed to the asymmetrical stretching vibration of the free carboxylic groups.³⁴ Most other detected signals remained unaffected by the SW immersion.

When the p(TiPSA) coatings were immersed in SW, the cleavage of the triisopropylsilyl groups led to a successive change of the surface properties that started already several minutes after immersion (CBCA change in Figure 3b) and lasted up to 2 weeks (water uptake change in Figure 2b). Based on the different surface analysis experiments, five different preimmersion time points were selected for further investigation, and the different hydrolysis states of the p(TiPSA) coatings were challenged against two marine organisms in dynamic attachment assays. For both organisms, hydrophobic pBMA was included as nonimmersed, negative reference (Figure 5a and c). The highest organism density in the diatom accumulation assay was observed for pBMA in all cases (1383 cells/ mm^2 , 1222 cells/ mm^2 , and 853 cells/ mm^2). Moreover, the preimmersion of the coatings improved their resistance properties and reduced the density of attached diatoms. The

diatom density decreased significantly ($p < 0.05$) for the silyl acrylate-based coatings that were preimmersed for 3 h and 1 day in SW compared to the pristine coating. For the pristine silyl acrylate-based coatings, a reduced density of 52% was observed compared to pBMA. When preimmersing the coatings for 3 h, the density decreased by 83% compared to the negative controls. The coatings, which were immersed for 1, 7, and 14 d, exhibited the lowest diatom density, with only 10% compared to pBMA. Interestingly, prolonged preimmersion for more than 24 h in SW did not lead to a significant improvement in the performance against *Amphora* sp. For the bacterial attachment assay, the highest organism density was observed on hydrophobic pBMA (3143 cells/ mm^2 , 792 cells/ mm^2 , and 1019 cells/ mm^2). The organism density decreased significantly ($p < 0.05$) after a preimmersion duration of 1 and 7 d compared to the pristine silyl acrylate-based coating. The pristine TiPSA-based coatings exhibited a bacterial density of 76% compared to the negative control, while preimmersing the coatings for 3 h did not lead to a significant improvement. The coatings that were preimmersed for 1 day in SW showed a significant reduction of 73% compared to the pristine coatings. The coatings with the highest hydrolysis state (7 and 14 d) exhibited the lowest bacterial density of 9% and 5% compared to pBMA, respectively. Thus, longer preimmersion periods of 7 days did not lead to significant changes in the AF performance against *C. marina*. Figure 5 b and d show that plenty of agglomerates were observed on the hydrophobic pBMA, while fewer agglomerates were found on the preimmersed silyl acrylate coatings. Furthermore, evenly distributed individual cells were detected on all silyl acrylate-based coatings.

Since laboratory experiments with one species of organism highlighted the importance of the hydrolysis states on the propensity of the coating to reduce attachment, we further investigated the coatings against a broad variety of species in short-term field exposure experiments at Port Canaveral (Florida) under both static and dynamic conditions for 3–4 days. First, the affinity of the different hydrolysis states against sediment incorporation was investigated for 60 min under dynamic conditions. PEGMA was included as hydrogel control, and a relative absorbance of 39.4 was measured. In comparison to the PEGMA hydrogel, all silyl acrylic coatings showed a relatively low silt uptake. The lowest silt accumulation was found for the pristine silyl acrylic coating (rel abs 3.9). With increasing preimmersion time, the silt accumulation increased, while the highest sediment accumulation was observed for the coatings that were preimmersed for 1 day (rel abs 11.3). Interestingly, longer preimmersion of the coatings for 7 days and 14 days did not lead to a significant decrease in sediment uptake (rel. abs. of 10.3 and 8.6, respectively). Yet, compared to the PEGMA hydrogel coatings, the silt uptake was more than 70% lower on the silyl acrylates, showing in all cases only a weak affinity toward silt uptake.

After challenging the coatings against sediment uptake, the biomass accumulation on the different hydrolysis states was examined by short-term field exposure experiments. For the static immersion, the samples were vertically arranged in a frame and immersed at a depth of 0.5 m.⁴⁵ In Figure 6b and c, the average biomass density as determined by fluorescence microscopy is displayed as fluorescence coverage. For the static exposure experiments, the fluorescence coverage decreased significantly for the silyl acrylate-based coatings that were preimmersed for 3 h, 1 d, and 7 d (Figure 6b). The highest fluorescence coverage was observed for the pristine p(TiPSA)

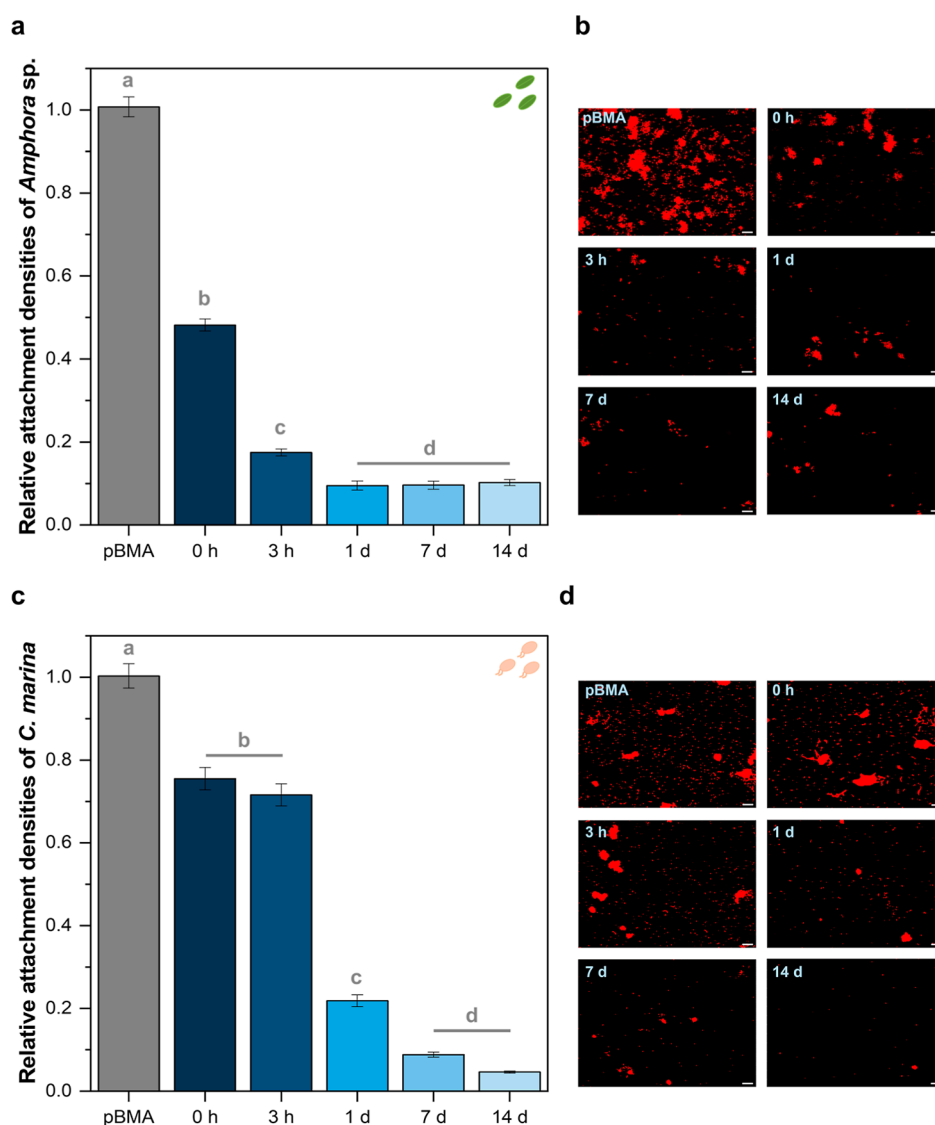


Figure 5. Colonization of the p(TiPSA) coatings at different hydrolysis states by the marine diatom *Amphora* sp. and the bacterium *Cobetia marina* under dynamic conditions. (a) Relative attachment densities ($n = 3$) of *Amphora* sp. normalized to hydrophobic pBMA with values of 1383 cells/mm², 1222 cells/mm², and 853 cells/mm². The assays were conducted for 60 min using a diatom concentration of 2 Mio cells/ml. Letters a–d mark bars between which differences were statistically significant ($\alpha = 0.05$). (b) Fluorescence images of attached *Amphora* sp. on surfaces with different hydrolysis states. Scale bars equal 50 μ m. (c) Relative attachment densities ($n = 3$) of *C. marina* normalized to pBMA with values of 3143 cells/mm², 792 cells/mm², and 1019 cells/mm². The assay was conducted for 45 min using a bacterial concentration of 7.5 Mio cells/ml. Bars sharing different letters a–d indicate statistically significant differences ($\alpha = 0.05$). (d) Fluorescence images of attached *C. marina* on coatings with different hydrolysis states. Scale bars equal 50 μ m.

coatings with 4.5%. After 7 days of preimmersion, the fluorescence coverage decreased to 1.5%, and no significant change in the fluorescence coverage was observed for a longer preimmersion duration. The hydrophobic pBMA controls experienced a fluorescence coverage of 0.6%, which was significantly lower than the coverage found on the silyl acrylate-based coatings.

Dynamic immersion tests were conducted at the FIT field test site using a custom-built rotating disk setup with a rotation speed of 13 rpm, and the samples were immersed at a depth of 0.5 m.⁴⁵ The y-scale of the plot in Figure 6c was chosen so that the pBMA fluorescence coverage bar had the same height as for the pBMA control of the static field experiments in Figure 6b for better comparison. For the dynamic experiments, no specific trend was observed when comparing the different hydrolysis states, which seems at first sight surprising. In

comparison, the overall density of the silyl acrylate-based coatings was rather comparable to longer preimmersed samples in the static experiments.

DISCUSSION

In this work, homogeneous TiPSA-based polymers using an EGDA cross-linker were prepared via a free-radical grafting-through polymerization approach to study the influence of different hydrolysis states on the antifouling performance under both laboratory and short-term field conditions. For most of the commercially available paints, a mixture of various monomers is usually used, where the content of the silyl (meth)acrylates accounts for more than 50 wt %. We fabricated a polymer solely based on the hydrolyzable TiPSA in order to investigate its inherent hydrolysis properties with a focus on its consequences for the antifouling performance.

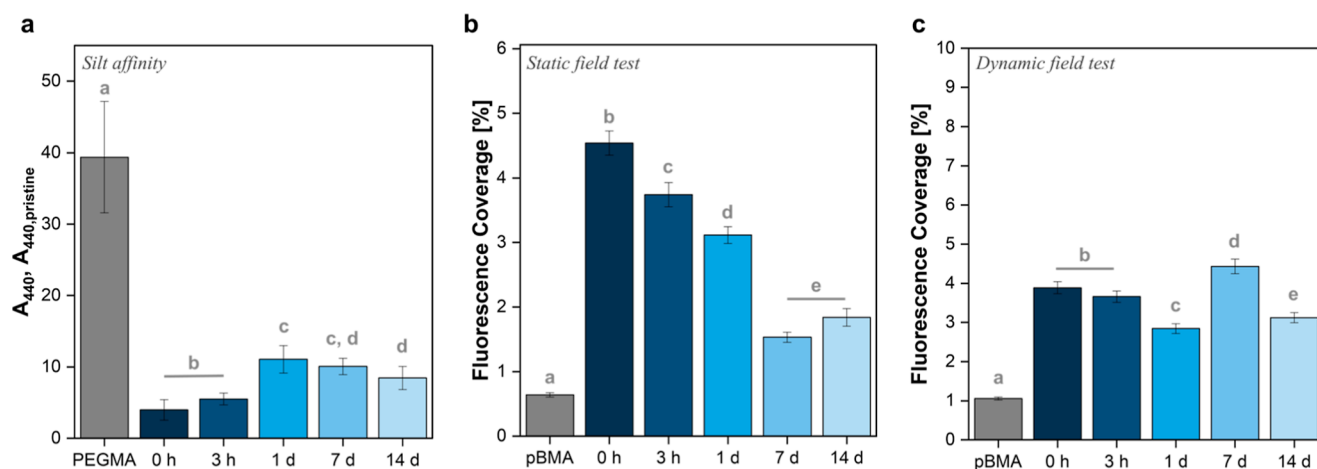


Figure 6. Silt adsorption susceptibility and short-term field immersion assays (a) Silt incorporation into the different preimmersed coatings after exposure for 60 min under dynamic conditions. The relative absorbance was determined by measuring the absorbance before and after silt incubation. (b,c) Short-term field exposure of the coatings was done at Port Canaveral, and the average organism density on the different coatings that were immersed under (b) static and (c) dynamic conditions for 3–4 days is shown.

EGDA was added as a cross-linker to restrict any ablation and limit the study only to the hydrolysis of the external surface layer. Success of the polymerization was monitored using ATR-FTIR spectroscopy. We were able to show that the C=C bond, which can be observed at 1620 cm^{-1} , disappeared in the TiPSA-based polymer, indicating successful polymerization via the grafting-through approach.⁴⁷ The pristine, nonimmersed coating was characterized by a coating weight and thickness of approximately 130 mg and $150\text{ }\mu\text{m}$, and static water contact angles in air of 97° revealed hydrophobic surface properties.

As intended by the use of the EGDA cross-linker, a high stability with an RMF > 98% and a moderate swelling of 23% were observed for the coatings in SW over a period of 14 days, suggesting a certain hydrogel character due to the transformation of the triisopropylsilyl pendant group, which resulted in an increased hydrophilicity over time. With longer preimmersion, altered mechanical properties of the bulk were observed as the coatings became significantly softer (Figure 2c). Interestingly, Dai et al. reported that the pencil hardness of an MMA/TiPSA (85/15) copolymer remained constant at 2H when immersing the coatings for 28 days in ASW, which suggests that the high content of MMA in the coatings decreased the swellability and thus enhanced the long-term mechanical stability.³⁴ The static WCA in our experiments decreased with an increased immersion duration in SW under dynamic conditions until it equilibrated after 7 days, revealing that the progressing hydrolysis that involves the transformation of the pendant groups at the external surface layer reached an equilibrated hydrolyzed state. In situ characterization of the hydrolytic behavior by captive bubble contact angle measurements showed that within the second of immersion, the coatings became more hydrophilic, which is due to reorientation effects of the acrylate units toward the surface, leading to decreased CBCAs immediately after submersion. With an increasing incubation time, the surface properties are governed by the hydrolysis of the triisopropylsilyl pendant groups at the external surface layer, which resulted in the formation of carboxyl groups (Figure 3b). Interestingly, after an immersion for 1 day, the coatings showed CBCAs of approximately 24° , and longer immersion only resulted in minor changes, suggesting that under water, a surface fully saturated with carboxylate groups is formed already after 1 day

of immersion. In general, we noted a larger contact angle in the WCA experiments, which might be due to the drying of the samples before the measurement. It could be that during this drying process, some of the hydrophilic, high surface-energy groups reorient into the film and are not as exposed as in the CBCA measurements, in which the samples were under water the entire time. Similar observations of declining static water contact angles that level off after several days of immersion were already described before for silyl (meth)acrylate-containing copolymers, and it was reported that the required equilibration time depends on the exact composition of the coatings.^{34,35,49} The slowly succeeding hydrolysis was also visible in the ATR-FTIR measurements as the broad OH peak ($\sim 3680\text{--}3110\text{ cm}^{-1}$) as well as the carboxylic peak (1570 cm^{-1}) increased with increasing immersion duration of the silyl acrylic coatings (Figure 4).^{34,47}

Laboratory organism assays that challenge the coatings against the attachment of the diatom *Amphora* sp. and the bacterium *C. marina* showed a strong influence of the respective hydrolyzed surface state and its antifouling performance. In both cases, the highest organism density was observed for the hydrophobic pBMA reference. As reported earlier, diatoms readily colonize hydrophobic surfaces.⁵⁰ The pristine, nonimmersed coatings already revealed a reduced adhesion of the diatoms and bacteria by 52% and 24% relative to pBMA, respectively. A hydrolyzed surface state, which has formed after 1 day of preimmersion, reduced the adhesion of both organisms and thus strongly enhanced the AF performance. Longer incubation might lead to even higher hydrolysis states, but no substantial further improvements in the antifouling performance against diatoms were found. The bacteria seemed to react more sensitively to conversions occurring during external surface hydrolysis. Such organism-dependency was already described by Dai et al., who reported a 39% decreased density for the diatom *N. incerta* settled under static conditions on coatings containing MMA and low amounts of TiPSA (85/15). The coatings were preimmersed for 3 days prior to the organism experiment and compared to the pristine, non-immersed surfaces. The density of the bacterium *Pseudomonas* sp. when allowed to attach under static conditions was not affected by the preimmersion.³⁴ The general notion that preimmersed coatings show good antifouling properties agrees

well with studies that tested coatings with a trialkylsilyl methacrylate content of approximately 30 mol %, which showed both a reduction in *Navicula salinicola* attachment and proliferation.⁵¹ In addition to the laboratory assays, we tested our coatings with different hydrolysis states in a real mixed-species environment under static and dynamic conditions for 3–4 days at Port Canaveral.⁵² For both immersions, the pBMA reference revealed the lowest fluorescence coverage compared to the silyl acrylate coatings. A good performance of the pBMA under dynamic field conditions was already described previously, and it was noted that the samples showed only a very low tendency for sediment accumulation.^{13,53} The fluorescence coverage on the silyl acrylic coatings decreased with a progressing hydrolysis state under static conditions, which agreed very well with the laboratory experiments. Thus, 7 days of preimmersion revealed the best AF performance, and longer immersion durations did not further improve the performance. The dynamic experiment revealed similar coverages for the different hydrolysis states. We can only speculate that the dynamic conditions in an environment that contains suspended particles led to an accelerated hydrolysis and removal of the triisopropylsilyl pendant group and thus a quicker transformation into a hydrophilic interface. For the real situation on a ship hull, this might imply that vessels starting to cruise at a certain velocity immediately after immersion of a fresh silyl (meth)acrylate-containing coating might be able to quickly establish a hydrophilic, low-fouling condition of the surface. Furthermore, the very low affinity of the hydrolyzed coatings to accumulate silt (Figure 6a) seems to be important for their excellent performance in short-term field exposures. The addition of amino-rich siloxanes into (meth)acrylate-based coatings could be an additional approach to further control the polishing rates for optimized coating performance, as recently described.⁵⁴ Overall, a hydrolyzed surface state seems to be beneficial for an enhanced resistance of the coatings against different marine organisms, as a longer water contact time results in optimized hydrolytic properties, improved water uptake, and a decreased elastic modulus.

CONCLUSION

In summary, a polymeric coating that solely contains TiPSA was synthesized using a photoinduced, free-radical grafting-through polymerization approach. For fabrication, the coatings were cross-linked, which enabled the study of the in situ behavior at the coating-water interface. Upon water contact, rapid reorientation of chemical groups toward the surface and the successive hydrolysis were followed by WCA and CBCA goniometry as well as IR measurements. The progressing hydrolysis over time resulted in a hydrogel character as the swelling and softness of the coatings increased, while the stability remained nearly unaffected. Laboratory and short-term field exposure revealed that an advanced hydrolysis state can beneficially affect the antifouling properties of such coatings. The study further underlines the unique properties of silyl acrylate building blocks and demonstrates the importance of the hydrolysis state of the coatings with relevance for both the initial phase after immersion of a fresh coating into the ocean, as well as potential release mechanisms of added biocides.

ASSOCIATED CONTENT

Data Availability Statement

Data can be provided by the authors upon request.

AUTHOR INFORMATION

Corresponding Author

Axel Rosenhahn – Analytical Chemistry – Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0001-9393-7190; Email: axel.rosenhahn@rub.de

Authors

Louisa Vogler – Analytical Chemistry – Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0001-9827-7961

Regina Kopeck – Analytical Chemistry – Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany

Onur Özcan – Analytical Chemistry – Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0003-4947-2757

Kelli Hunsucker – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Cierra Braga – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Tim Patschorke – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Dylan Thomas Eggers – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Geoffrey Swain – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsami.5c25268>

Notes

The authors declare no competing financial interest.

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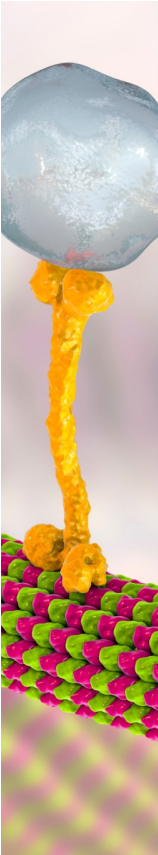
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